WireMod Spline Generation Guide for SBND

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Contents

1 Code Repositories

The spline generation workflow is currently split across two local repositories:

- `SBNDWireModSplineGeneration`: Python workflow scripts, ROOT output files, CSV summaries, and diagnostic plots.
- `WireMod-Spline-Generation-Guide`: this documentation project.

The current 2D workflow uses PyROOT for all ROOT I/O and histogram projections, while Matplotlib is used for diagnostic plots. The ROOT environment is loaded through the wrapper script:

```
./scripts/run_with_root.sh <python script> <options>
```

2 Simple 2D Spline Generation

2.1 Goal

The first 2D spline production tests use lower-dimensional Gen2 histograms where the first two THnSparse axes are WireMod coordinates or angles, and the third axis is the hit-level response variable. The current samples are:

- `YZQ`: adaptive bins in (y, z), hit response axis Q.
- `YZW`: same (y, z) binning as `YZQ`, hit response axis W.
- `XTXZQ`: adaptive bins in (x, θ_{xw}), hit response axis Q.
- `XTXZW`: same (x, θ_{xw}) binning as `XTXZQ`, hit response axis W.

For each plane, the workflow constructs adaptive 2D regions from the data track-count histogram, extracts data and simulation hit-response distributions in each region, estimates the most probable hit-response scale with an iterative truncated mean, and stores the ratio

$$R_{\mathrm{WireMod}}(a, b) = \frac{\mathrm{MPV}_{\mathrm{data}}(a, b)}{\mathrm{MPV}_{\mathrm{MC}}(a, b)} \tag{1}$$

as a `TGraph2DErrors`. Here (a, b) denotes the two WireMod dimensions used for the file, for example (y, z) or (x, θ_{xw}).

2.2 Input Files

The Gen2 2D input files are stored in:

```
/Users/alexanderantonakis/Documents/Codex/2026-06-08/Gen2_2D
```

Each file contains one `hTrackN` and one `hHitN` object for each of the six wire planes. The `hTrackN` histograms define the binning statistics. The corresponding `hHitN` histograms provide the hit-response distributions used to extract the data/MC ratio.

2.3 Adaptive Binning

The adaptive binning is produced with `scripts/make_adaptive_binning_2d.py`. The default target is currently 5×10^5 tracks per 2D region. The script projects the 3D `hTrackN` sparse onto axes 0 and 1, converts the projection into a NumPy count matrix, and recursively partitions this plane.

The recursive splitting rule is:

1. Start from the full active 2D plane.
2. Try a quadrant split. If all four quadrants exceed the target count threshold, keep splitting each quadrant.
3. If the quadrant split is not allowed by the threshold, try a binary split along the selected fallback axis.

4. Keep the split only if both children exceed the threshold.
5. Otherwise, keep the current region as a terminal adaptive bin.

For `XTXZ` files, one side of the detector is mostly empty for each plane. The workflow therefore applies an active-x selection before adaptive splitting. The default `auto` behavior uses $x < 0$ for planes 0–2 and $x > 0$ for planes 3–5. This prevents the empty side of the histogram from forcing the algorithm into one overly large bin.

A typical command is:

```
./scripts/run_with_root.sh scripts/make_adaptive_binning_2d.py \
  --input ../Gen2_2D/Wire_Gen2_DATA_YZQ_26Aug2026_Full.root \
  --hist hTrack0 \
  --target 500000 \
  --max-plots 1
```

For the width response files, the same binning is reused from the corresponding charge file: `YZQ` binning is used for `YZW`, and `XTXZQ` binning is used for `XTXZW`.

Table 1: Adaptive 2D region counts from the current Gen2 data track histograms. The `XTXZQ` counts include the active-x side selection.

Plane	`YZQ` regions	`XTXZQ` regions
0	2145	826
1	2129	796
2	1056	913
3	2193	852
4	2140	862
5	1116	912

2.4 Ratio Extraction

The ratio extraction is performed with `scripts/make_2d_ratio_splines.py`. For each adaptive region, the script applies the region as an axis-0/axis-1 selection on the data and MC `hHitN` sparse histograms, projects the response axis, and computes the iterative truncated mean for the data and MC projected distributions.

The truncated-mean settings currently match the earlier 3D spline workflow:

- lower truncation: -2.0 RMS from the current mean,
- upper truncation: $+1.75$ RMS from the current mean,
- convergence tolerance: 10^{-4},
- maximum iterations: 100.

The central spline point is placed at the geometric center of each adaptive region. By default, the script also duplicates points at boundary edges and corners. These edge-augmented points help constrain interpolation near the boundary of the valid 2D phase space. The output ROOT file contains:

- `data_mpv_N`: data truncated-mean MPV graph,
- `mc_mpv_N`: MC truncated-mean MPV graph,
- `splines_N`: data/MC ratio graph.

A typical ratio command is:

```
./scripts/run_with_root.sh scripts/make_2d_ratio_splines.py \
  --data ../Gen2_2D/Wire_Gen2_DATA_YZQ_26Aug2026_Full.root \
  --mc ../Gen2_2D/Wire_Gen2_MC_YZQ_26Aug2026_Full.root \
  --hit-hist hHit0 \
  --idx 0 \
  --binning-label YZQ_26Aug2026
```

For `YZW`, the output label and binning label are different:

```
./scripts/run_with_root.sh scripts/make_2d_ratio_splines.py \
  --data .../Wire_Gen2_DATA_YZW_26Aug2026_Full.root \
  --mc .../Wire_Gen2_MC_YZW_26Aug2026_Full.root \
  --hit-hist hHit0 \
  --idx 0 \
  --output-label YZW_26Aug2026 \
  --binning-label YZQ_26Aug2026
```

2.5 Current Outputs

The current output files are written under:

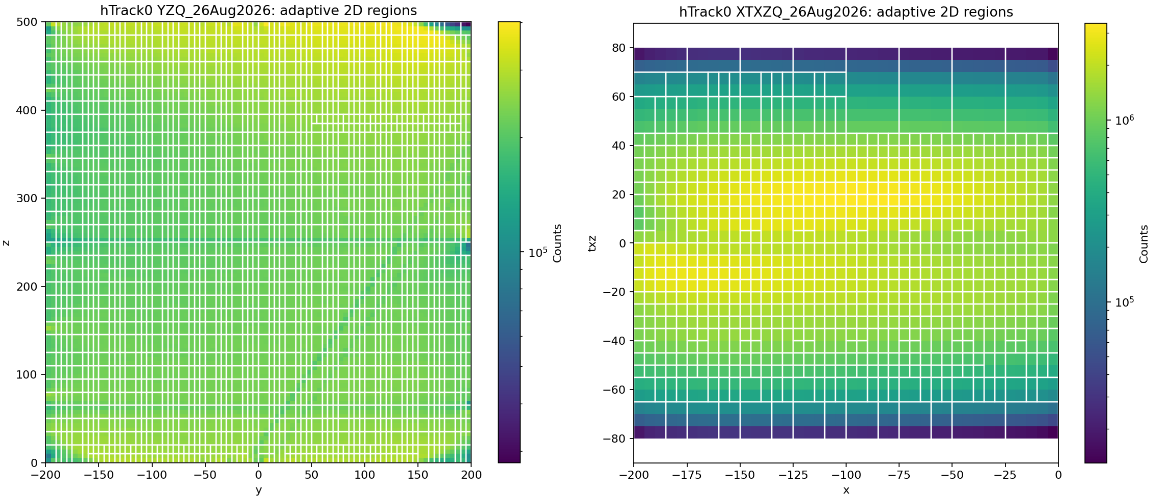
```
plots/adaptive_2d
plots/ratio_splines_2d
```

in the `SBNDWireModSplineGeneration` repository. CSV files store the adaptive regions and MPV ratio summaries, while ROOT files store the final graphs used downstream by WireMod.

Table 2: Current 2D ratio spline point counts. Each entry gives central adaptive-region fits followed by the number of graph points after edge augmentation.

Plane	`YZQ`	`YZW`	`XTXZQ`	`XTXZW`
0	2145 / 2350	2145 / 2350	826 / 897	826 / 897
1	2129 / 2336	2129 / 2336	796 / 864	796 / 864
2	1056 / 1209	1056 / 1209	913 / 996	913 / 996
3	2193 / 2404	2193 / 2404	852 / 920	852 / 920
4	2140 / 2346	2140 / 2346	862 / 930	862 / 930
5	1116 / 1277	1116 / 1277	912 / 990	912 / 990

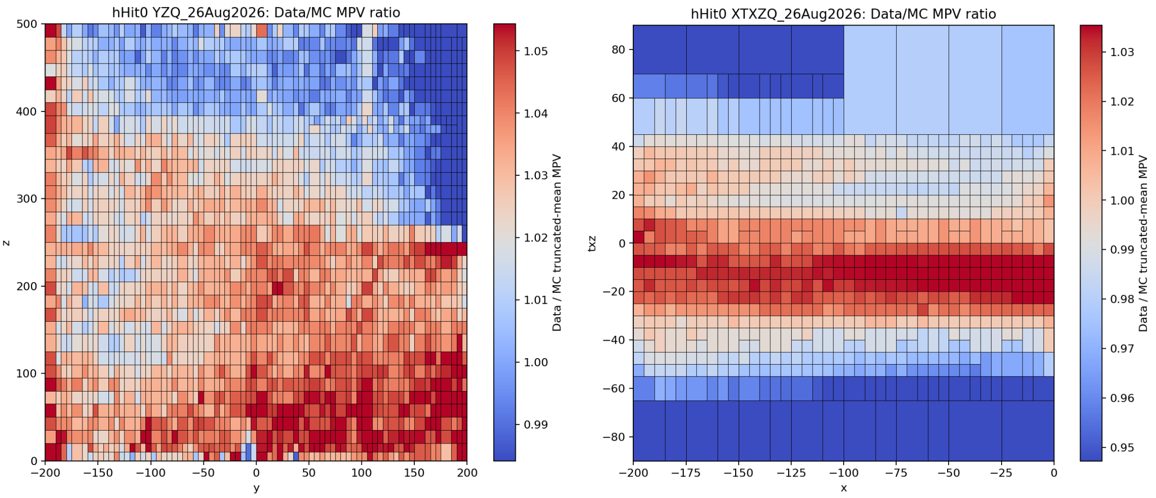
Figures 1 and 2 show representative outputs for plane 0. The full plane-by-plane diagnostic set is included in Section A.



(a) `YZQ` adaptive regions.

(b) `XTXZQ` adaptive regions.

Figure 1: Example adaptive 2D binnings for plane 0. The `XTXZQ` binning uses the active $x < 0$ side for plane 0.



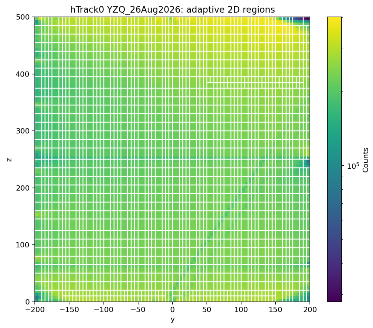
(a) `YZQ` ratio map.

(b) `XTXZQ` ratio map.

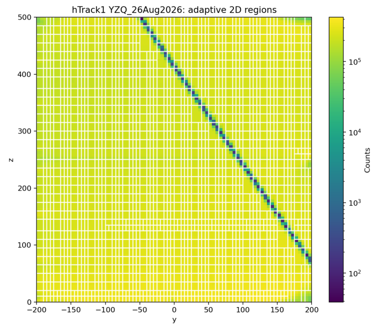
Figure 2: Example data/MC truncated-mean MPV ratio maps for plane 0.

Appendix A 2D Splines in Gen2

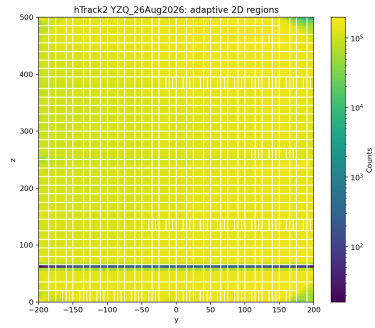
A.1 Adaptive Binning Diagnostics



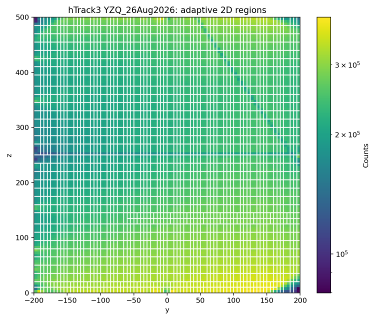
(a) Plane 0



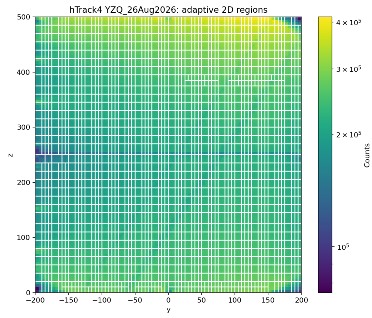
(b) Plane 1



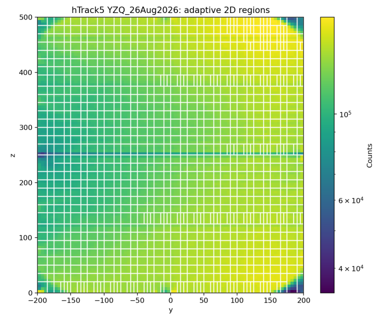
(c) Plane 2



(d) Plane 3

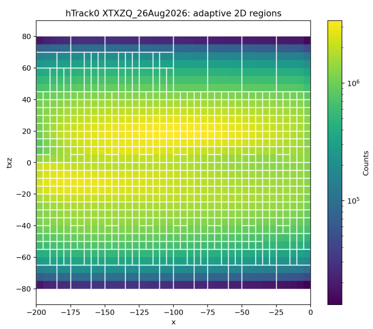


(e) Plane 4

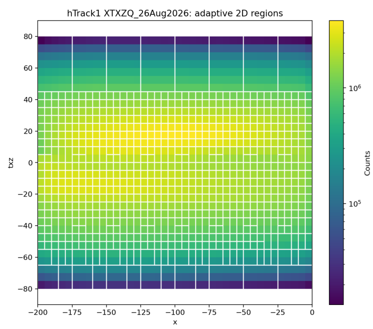


(f) Plane 5

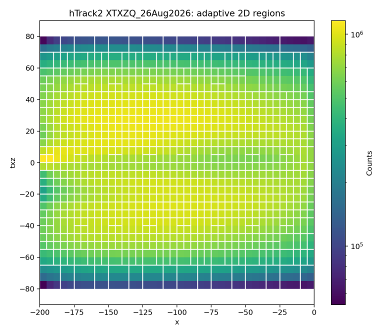
Figure 3: `YZQ` adaptive 2D regions for all planes.



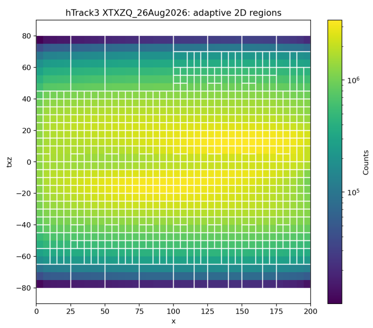
(a) Plane 0



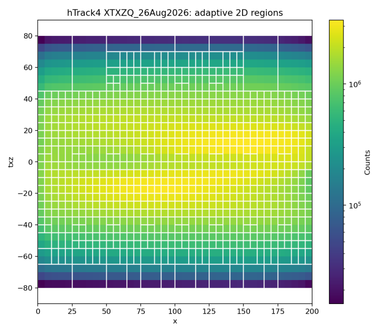
(b) Plane 1



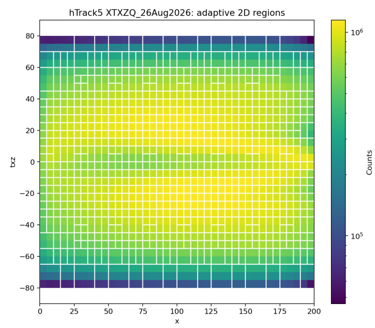
(c) Plane 2



(d) Plane 3



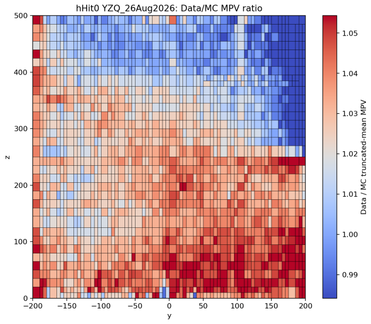
(e) Plane 4



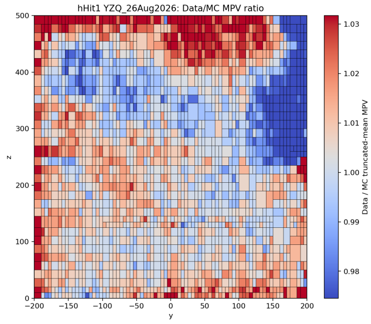
(f) Plane 5

Figure 4: `XTXZQ` adaptive 2D regions for all planes.

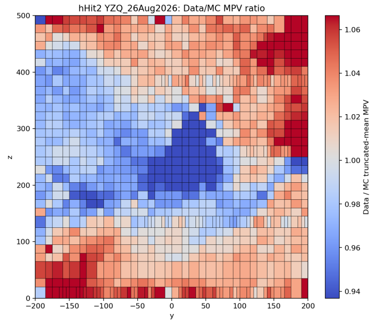
A.2 Ratio Map Diagnostics



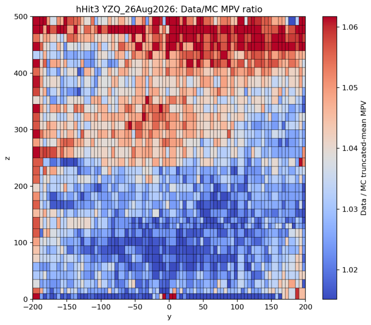
(a) Plane 0



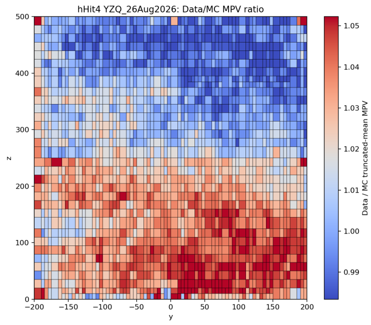
(b) Plane 1



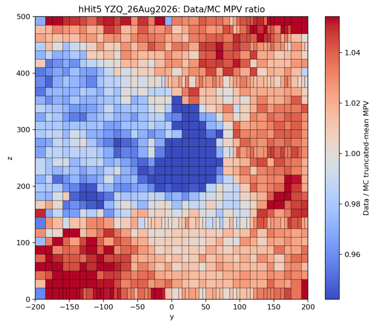
(c) Plane 2



(d) Plane 3

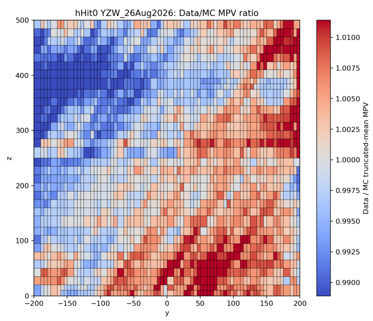


(e) Plane 4

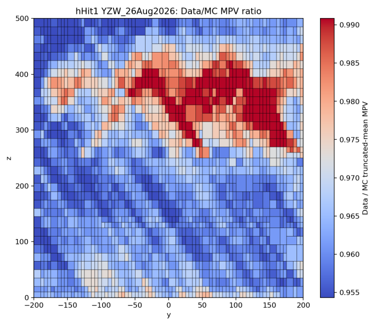


(f) Plane 5

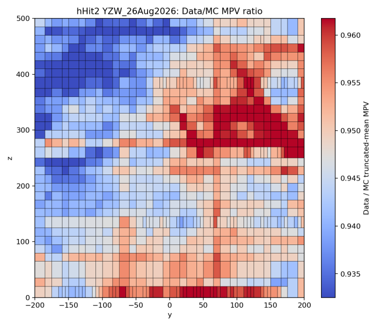
Figure 5: `YZQ` data/MC MPV ratio maps for all planes.



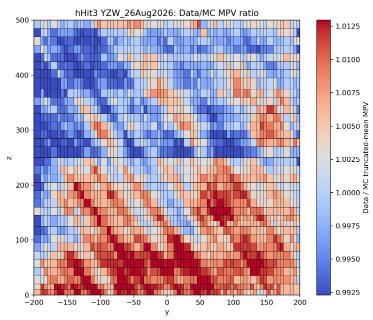
(a) Plane 0



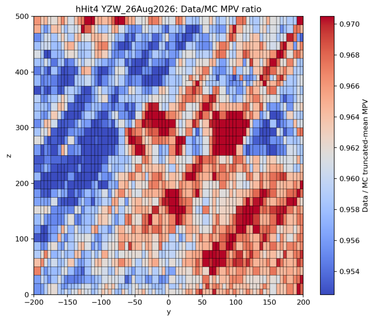
(b) Plane 1



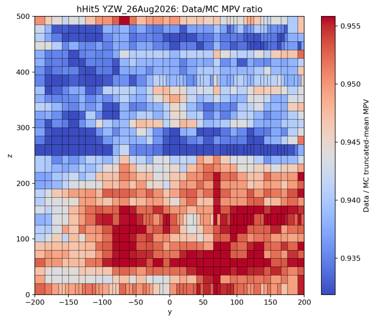
(c) Plane 2



(d) Plane 3

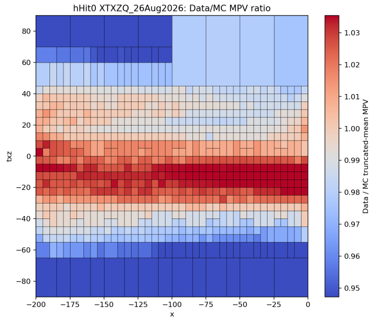


(e) Plane 4

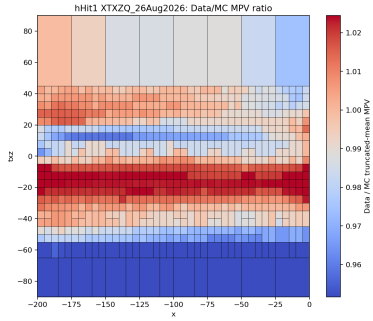


(f) Plane 5

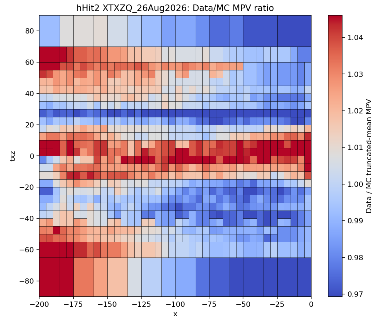
Figure 6: `YZW` data/MC MPV ratio maps for all planes.



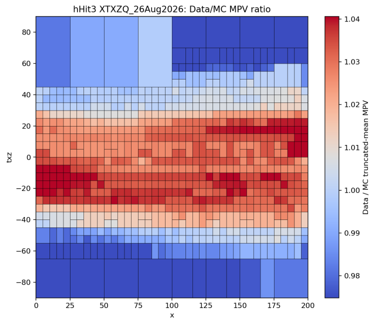
(a) Plane 0



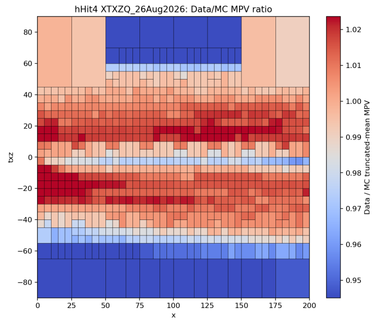
(b) Plane 1



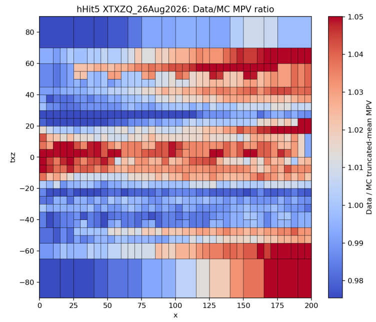
(c) Plane 2



(d) Plane 3

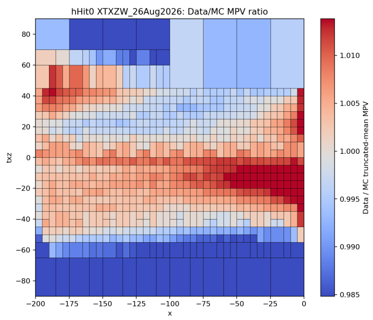


(e) Plane 4

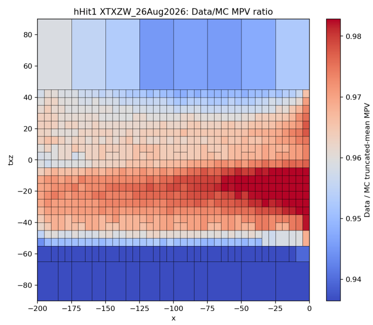


(f) Plane 5

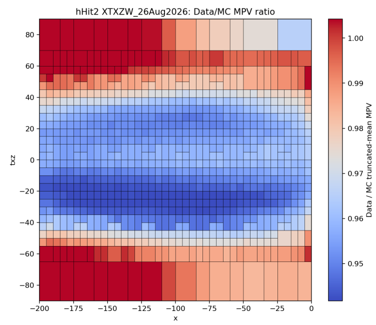
Figure 7: `XTXZQ` data/MC MPV ratio maps for all planes.



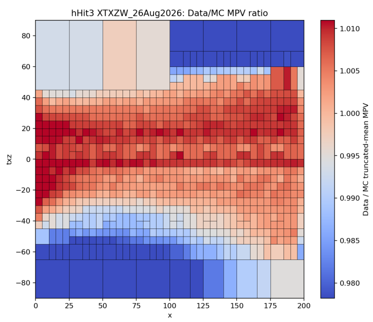
(a) Plane 0



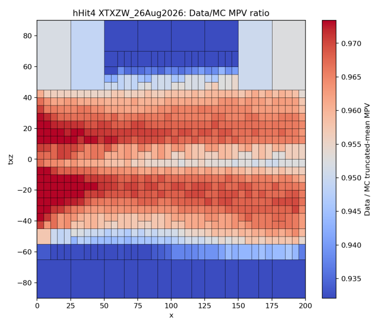
(b) Plane 1



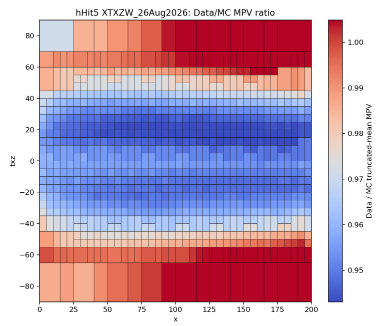
(c) Plane 2



(d) Plane 3



(e) Plane 4



(f) Plane 5

Figure 8: `XTXZW` data/MC MPV ratio maps for all planes.